Wentao Zhang

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Education

Southeast University 2021/09 - 2025/09
Ph.D. in Electrical Engineering Nanjing, Jiangsu, China

- Dissertation: "Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles".
- **Industrial prototype development with Inovance** (world-leading EV traction machine supplier): technical lead for two high-speed oil-cooled PM traction machines taken through the full design-build-test cycle. **20,000 rpm, A-class:** 270 kW, 380 Nm, 7.87 kW/kg, 97.4% peak efficiency. **18,000 rpm, A0-class:** 113 kW, 138 Nm, 7.78 kW/kg, 95.3% peak efficiency (all measured).
- **Design:** multi-objective optimisation coupling fast index models, electromagnetic FEA, thermal networks and stress evaluation. **Manufacture:** engineering drawing and tolerance release; coordination of external suppliers for machining, lamination stacking, hairpin forming and assembly. **Test:** AVL dynamometer characterisation (efficiency maps, thermocouple and waveform data) used to validate the design models.
- 2025 MDPI *Machines* Best PhD Thesis Award (awarded for this doctoral thesis).

Southeast University 2018/09 - 2021/06
M.Eng. in Electrical Engineering Nanjing, Jiangsu, China

- Dissertation: "Analysis and Design of High-Performance Wound-Excited Magnetic Field Modulation-based Brushless Machines".
- 1st Prize, IEEE IAS CMD Thesis Contest, 2023, and Outstanding Master Degree Thesis Award, Jiangsu Province, 2022 (both for this thesis).

Southeast University 2014/09 - 2018/06
B.Eng. in Electrical Engineering Nanjing, Jiangsu, China

- Dissertation: "Comparative Study of Control Systems for PM Machine in Electric Vehicles".

Work Experience

University of Nottingham 2025/09 - present
Research Assistant, Power Electronics and Machines Centre (PEMC) Nottingham, United Kingdom

- Working on next-generation magnet-free synchronous machines within the EPSRC-funded Sustainable Multi-Sector Electrification Using Advanced Integrated Motor Drive Technologies (AIMD) programme (Nottingham, Bristol, Newcastle).
- Developing an **in-house finite element code package** in Python for electromagnetic, structural-modal and mechanical-stress analysis of electrical machines, including geometry and mesh generation, solver and post-processing.
- Extending **physics-informed neural surrogates** to replace expensive FEA in design-space exploration.
- Leading a project on **LLM-agent-assisted design and training of physics-informed surrogates** (PINNs, Transformers).

Research Profile

- **Area.** Physics-informed AI for the design and validation of electrical machines and drives, the energy converters behind electric transport, renewable generation and robotics.
- **Connection to AI.** *Build:* field-level deep-learning surrogates trained on my own finite element solver [C1]. *Leverage:* hybrid AI-plus-physics design optimisation [J6], [P3] and digital twins checked against prototype measurements [J10], [J11]. *Understand:* where LLM agents running engineering simulations break physics or disagree with measurement (current project).
- **Relative to opportunity.** PhD awarded September 2025; the record below (24 journal papers, 20 in IEEE Transactions, 11 as first author) was built over doctoral study and one year of postdoctoral employment.

Research Funding

- **N-ZEEE Flexible Funding award** (Net-Zero Enabling Electrical Engineering network, 2026), **£2,700**, approved by the network's selection panel on my proposal, to present [C2] at IEEE ICEMS 2026 (Singapore, 24 to 27 November 2026) and attend its tutorial day.
- **OpenAI Researcher Access Program** (2026): in-kind API credits for the LLM-agent surrogate project I lead.
- **Participating researcher** on externally funded programmes: the EPSRC **AIMD** programme at Nottingham (2025 to present) and the **Inovance**-funded traction-machine development programme at Southeast University (2021 to 2025).

Research Plan and Funding Strategy (first five years)

Aim: AI models of electrical machines and drives that engineers can trust to make decisions, because each learned model is tied to physics and checked against measurement from the built machine.

- **Theme 1, building physics-informed field surrogates (Build AI).** Extend the surrogate [C1] to operator learning across topologies, with uncertainty quantification and active learning.
- **Theme 2, measurement-calibrated digital twins (Leverage AI).** Couple the surrogate with the validated thermal [J10], [J12] and mechanical [J11] models, and update the twin from test-bench and in-service data for derating, condition monitoring and life prediction in electrified mining, heavy transport and renewable generation.
- **Theme 3, AI agents for engineering design, and where they fail (Understand AI).** Benchmarks of where simulation-running agents break physics or disagree with measurement.
- **Funding route.** Year 1: **ARC DECRA** (eligible, PhD 2025) and University start-up support. Years 1 to 3: **ARC Linkage** and **CRC-P** projects with industry in electrified mining, transport and energy. Years 3 to 5: **ARC Discovery** and industry co-funded PhD scholarships, with links to the Net Zero Institute and the Australian Centre for Robotics.

Teaching and Supervision

- **Teaching assistant**, SEU postgraduate course *Integrated Practice of Electrical Machine Systems* (2021 to 2025): tutorial preparation and delivery, Q&A and consultation hours, structured written feedback on student final-project proposals, and mark recommendations; developed tutorial and assessment materials and research-derived resources (a coupled electromagnetic-thermal modelling exercise and an embedded motor-drive practical).
- **Informal supervision** of junior PhD and MSc colleagues in machine modelling, FEA and thermal networks.

Professional Activities

- **Referee** for IEEE Transactions journals, including *IEEE Trans. Industry Applications*, *IEEE Trans. Energy Conversion* and *IEEE Trans. Transportation Electrification.*
- **Co-Organiser and Co-Chair**, ICEM 2026 Special Session "Next-Generation Electrified Transport: Advances in Electric Drive Systems", Funchal, Portugal, September 2026 (with Dr Lingyun Shao, NUAA).

Awards and Honours

- 2025 **1st Prize, IEEE TEC Electric Vehicle Design Competition**.
- 2025 **2nd Prize Paper Award, IEEE Transactions on Transportation Electrification** (for first-author paper [J5]).
- 2025 **MDPI *Machines* Best PhD Thesis Award**.
- 2024 2nd Prize, 11th Energy Equipment Innovative Design Competition for China Postgraduate.
- 2023 CIEEC Best Paper Award.
- 2023 1st Prize, IEEE IAS CMD Thesis Contest.
- 2023 Wiscom System Scholarship.
- 2023 1st Prize Academic Scholarship.
- 2022 Outstanding Master Degree Thesis Award, Jiangsu Province.

Research and Technical Expertise

- **Physics-informed and hybrid AI.** Full-field magnetic surrogate [C1]: R^2 = 0.9996 (torque) and 0.9730 (iron loss) on unseen designs, **more than 94% faster** than FEA. Hybrid optimisation cutting 67 h of FEA to 2.3 h [J6]; **dual-model** database patent [P3].
- **Validated multi-physics digital twins.** Oil-cooled thermal network **validated against thermocouples across the full operating map** [J10], [J12], [P4]; rotor stress models verified by a **22,500 rpm overspeed test** [J11]; PWM effects on loss and temperature [J22].

- **Design for manufacture.** *n*-layer flat-wire (hairpin) winding methodology built around forming limits [J5] (Prize Paper Award); glued versus welded stator cores and vibration [J23]; bridge-retained versus carbon-fibre-sleeved IPM rotors [C2].
- **2D/3D FEA.** In-house Python FE package (see Work Experience); 3D end-effect analysis in ANSYS Maxwell and JMAG [J9].
- **Machine design and test.** IPM, SPM, wound-field and hybrid-excitation machines; AC copper and PWM iron losses; AVL dynamometer testing (efficiency maps, thermocouple instrumentation, acoustics); design and commissioning of a **three-phase full-bridge converter** and its control.
- **Software.** Python, PyTorch, MATLAB/Simulink; ANSYS Maxwell 2D/3D, JMAG, FEMM 4.2; Motor-CAD, ANSYS Mechanical; SolidWorks, AutoCAD, SpaceClaim; C/C++ on TI C2000 and STM32. **Languages:** Mandarin Chinese (native), English (full professional proficiency).

Research Publications

24 journal papers (20 in IEEE Transactions, 11 as first author), 19 peer-reviewed conference papers and 9 granted patents.

Journal Papers (24 papers, 20 in IEEE Transactions)

[J1] **Wentao Zhang**, W. Hua, Z. Wu, G. Zhao, Y. Wang, and W. Xia, "Analysis of DC Winding Induced Voltage in Wound-Field Flux-Switching Machine With Air-Gap Field Modulation Principle," *IEEE Trans. Ind. Electron.*, vol. 69, no. 3, pp. 2300–2311, Mar. 2022.

[J2] **Wentao Zhang**, Z. Wu, W. Hua, Z. Zhu, Z. Wu, J. Hu, et al., "Reduction of Open-Circuit DC Winding Induced Voltage and Torque Pulsation in the Wound Field Switched Flux Machine by Stator Axial Pairing of Tooth Tips," *IEEE Trans. Ind. Appl.*, vol. 58, no. 2, pp. 1976–1990, Mar.–Apr. 2022.

[J3] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "A Stable and Computationally Efficient Spatial Harmonic Model for Predicting the DC Winding Induced Voltage in WFSF Machine," *IEEE Trans. Ind. Appl.*, vol. 59, no. 4, pp. 3966–3977, Jul.–Aug. 2023.

[J4] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "Analysis and Improvement of DC Winding Current Dynamics in Wound Field Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 60, no. 4, pp. 6023–6032, Jul.–Aug. 2024.

[J5] **Wentao Zhang**, Z. Wu and W. Hua, "**Design of n-Layer Flat Wire Winding in Permanent Magnet Machine for Electric Vehicles**," *IEEE Trans. Transport. Electrific.*, vol. 11, no. 3, pp. 7798–7810, 2025. (2025 IEEE TTE Prize Paper Award, 2nd Prize.)

[J6] **Wentao Zhang**, Z. Wu, L. Jin, Y. Fan, W. Hua, and M. Cheng, "**Analysis and Multi-Objective Optimization of the Hybrid Excitation Switched Flux Machine**," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 2996–3006, Mar.–Apr. 2025.

[J7] **Wentao Zhang**, Z. Wu, Z. Wang, W. Hua, Y. Fan, Z. Xu, et al., "Design Tradeoff Between Flux Regulation Capability and DC Winding Induced Voltage in Hybrid Excitation Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 61, no. 5, pp. 7198–7208, Sep. 2025.

[J8] **Wentao Zhang**, Y. Fan, Z. Zhu, Z. Wu, W. Hua, and M. Cheng, "Analysis of DC Winding Induced Voltage in Wound-Rotor Synchronous Machines by Using the Air-Gap Field Modulation Principle," *World Electr. Veh. J.*, vol. 13, no. 11, Art. no. 215, 2022.

[J9] **Wentao Zhang**, Z. Wu, Z. Cang, W. Hua, Y. Fan, and M. Cheng, "Influence of End-Effect in Switched Flux Machines," *IEEE Trans. Ind. Appl.*, vol. 62, no. 1, pp. 785–798, Jan. 2026.

[J10] **Wentao Zhang**, Z. Wu and W. Hua, "**High-Fidelity Lumped Parameter Thermal Model for Oil-Cooling Hairpin Winding PM Machines in EVs**," *IEEE Trans. Transport. Electrific.*, vol. 12, no. 2, pp. 2978–2991, Apr. 2026.

[J11] **Wentao Zhang**, Z. Wu and W. Hua, "**Design of Notching and Mechanical Stress Relieving Grooves for High-Speed PM Machine in EVs**," *IEEE Trans. Ind. Appl.*, early access, 2026.

[J12] Z. Wu, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "**A Transient-State Lumped Parameter Thermal Model for Brushless Wound Field Switched Flux Machines**," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 1, pp. 224–238, Mar. 2024.

[J13] Z. Wu, Z. Zhu, C. Wang, W. Hua, K. Wang, and **Wentao Zhang**, "Influence of Rotor Iron Bridge Position on DC-Winding-Induced Voltage in Wound Field Switched Flux Machine Having Partitioned Stators," *Chin. J. Electr. Eng.*, vol. 7, no. 3, pp. 20–28, Sep. 2021.

[J14] Z. Wu, Z. Zhu, S. Cai, W. Hua, and **Wentao Zhang**, "Enhancement of Torque Density in Wound Field Switched Flux Machines With Partitioned Stators Using Assisted Ferrites," *Chin. J. Electr. Eng.*, vol. 7, no. 3, pp. 42–51, Sep. 2021.

[J15] Z. Wu, L. Jin, W. Hua, **Wentao Zhang**, and M. Cheng, "Harmonics Orders Modeling of DC Winding Induced Voltage Pulsation in Wound Field Switched Flux Machines Under PWM Excitation," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 3, pp. 7040–7050, Sep. 2024.

[J16] Z. Wu, L. Jin, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "Influence of PWM Excitation on DC Winding Induced Voltage Pulsation in Wound Field Switched Flux Machines," *IEEE Trans. Ind. Appl.*, vol. 60, no. 1, pp. 460–476, Jan.–Feb. 2024.

[J17] Z. Wu, Z. Zhu, W. Hua, S. Akehurst, X. Zhu, **Wentao Zhang**, et al., "Analysis and Suppression of Induced Voltage Pulsation in DC Winding of Five-Phase Wound-Field Switched Flux Machines," *IEEE Trans. Energy Convers.*, vol. 34, no. 4, pp. 1890–1905, Dec. 2019.

[J18] L. Jin, W. Hua, U. B. Akuru, Z. Wu, **Wentao Zhang**, and M. Cheng, "Comparative Study on DC Winding Induced Voltage Pulsation of Wound Field Flux Modulation Machines Having Different Iron Core Structures," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 2874–2883, Mar.–Apr. 2025.

[J19] X. Yan, Z. Wu, W. Hua, **Wentao Zhang**, and M. Cheng, "A Mathematical Model for Wound Field Switched Flux Machine Considering Inductance Harmonics of Field and Armature Windings," *IEEE Trans. Ind. Appl.*, vol. 60, no. 5, pp. 6811–6822, Sep.–Oct. 2024.

[J20] H. Hua, X. Chen, D. Li, **Wentao Zhang**, Z. Wu, and W. Hua, "Investigation of Torque Improvement of Modular Stator Surface-Mounted Permanent Magnet Machines With Flux Gaps," *IEEE Trans. Ind. Appl.*, vol. 60, no. 5, pp. 6787–6798, Sep.–Oct. 2024.

[J21] J. Chen, W. Hua, L. Shao, Z. Wu, X. Yan, and **Wentao Zhang**, "Modified Magnetic Equivalent Circuit of Double-Stator Single-Rotor Axial Flux Permanent Magnet Machine Considering Stator Radial-End Flux-Leakage," *IET Electr. Power Appl.*, vol. 18, no. 2, pp. 195–207, Feb. 2024.

[J22] Z. Wu, X. Yan, Z. Wang, **Wentao Zhang**, and W. Hua, "**Influence of PWM Excitation on Loss and Temperature of Flat Wire Permanent Magnet Machines for Electric Vehicles**," *IEEE Trans. Ind. Electron.*, vol. 73, no. 4, pp. 5174–5185, Apr. 2026.

[J23] Z. Wu, E. Fu, J. Du, **Wentao Zhang**, and W. Hua, "**Quantitative Comparison of Electromagnetic Vibration of PMSMs With Glued and Welded Stator Core for Electric Vehicles**," *IEEE Trans. Ind. Appl.*, early access, 2025.

[J24] Z. Wu, **Wentao Zhang**, and W. Hua, "**Design, Analysis and Performance Evaluation of Hairpin Winding Permanent Magnet Machines**," *IEEE Trans. Transport. Electrific.*, accepted for publication, 2026.

Conference Papers (19 papers)

[C1] **Wentao Zhang**, Z. Wu and W. Hua, "**A Deep Learning Based Magnetic Field Surrogate Modeling Method for Interior Permanent Magnet Machine in EVs**," in *Proc. Int. Conf. Elect. Mach. (ICEM)*, Funchal, Madeira, Portugal, 2026. (Accepted.)

[C2] **Wentao Zhang**, Z. Wu and W. Hua, "Comparative Investigation of Bridge-Retained and Carbon Fiber Sleeved Interior Permanent Magnet Machines in Electric Vehicles," in *Proc. Int. Conf. Elect. Mach. Syst. (ICEMS)*, Singapore, 2026. (Accepted.)

[C3] **Wentao Zhang**, Z. Wu, W. Hua and Z. Zhu, "Reduction of Open-Circuit DC Winding Induced Voltage and Torque Pulsation in the Wound Field Switched Flux Machine by Stator Axial Pairing of Tooth-Tips," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, Gothenburg, Sweden, 2020, pp. 522–528.

[C4] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua and M. Cheng, "A Stable and Computationally Efficient Spatial Harmonic Model for Predicting the DC Winding Induced Voltage in WFSF Machine," in *Proc. IEEE Energy Convers. Congr. Expo.*, Detroit, MI, USA, 2022, pp. 1–7.

[C5] **Wentao Zhang**, Z. Wu, W. Hua, Y. Fan, Z. Xu and M. Cheng, "Design Tradeoff between Flux Regulation Capability and DC Winding Induced Voltage in Hybrid Excitation Switched Flux Machine," in *Proc. IEEE Energy Convers. Congr. Expo.*, Nashville, TN, USA, 2023, pp. 5060–5066.

[C6] **Wentao Zhang**, Z. Wu, L. Jin, Y. Fan, W. Hua and M. Cheng, "Analysis and Multi-Objective Optimization of the Hybrid Excitation Switched Flux Machine," in *Proc. Int. Elect. Energy Conf.*, Hefei, China, 2023, pp. 3371–3376. (Best Paper Award.)

[C7] **Wentao Zhang**, Z. Wu, W. Hua, Y. Fan and M. Cheng, "Influence of the End-Effect in Wound Field Switched Flux Machine," in *Proc. Int. Conf. Elect. Mach. Syst.*, Zhuhai, China, 2023, pp. 1223–1227.

[C8] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua and M. Cheng, "Analysis and Improvement of DC Winding Current Dynamics in Wound Field Switched Flux Machine," in *Proc. Int. Conf. Energy Technol. Future Grids*, Wollongong, Australia, 2023, pp. 1–6.

[C9] **Wentao Zhang**, Z. Wu, W. Hua and J. Qi, "Analysis of PM Volume Reduction Methods for Hybrid Excitation Switched Flux Machines," in *Proc. Int. Conf. Elect. Mach. Syst.*, Zhuhai, China, 2024, pp. 3152–3157.

[C10] **Wentao Zhang**, Z. Wu and W. Hua, "**Field-Oriented Design of Notching and Mechanical Stress Relieving Grooves for Rotors in Flat Wire Permanent Magnet Machines**," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, 2025, pp. 304–309.

[C11] Z. Wu, L. Jin, **Wentao Zhang**, Y. Fan, W. Hua and M. Cheng, "Influence of PWM Excitation on DC Winding Induced Voltage in Wound Field Switched Flux Machines," in *Proc. IEEE Energy Convers. Congr. Expo.*, Detroit, MI, USA, 2022, pp. 1–7.

[C12] L. Jin, W. Hua, U. B. Akuru, Z. Wu, **Wentao Zhang** and M. Cheng, "Comparative Study on DC Winding Induced Voltage Pulsation of Wound Field Flux Modulation Machines Having Different Iron Core Structures," in *Proc. Int. Conf. Elect. Mach. Syst.*, Zhuhai, China, 2023, pp. 442–447.

[C13] M. P. Koroma, W. Hua, Z. Wu, L. Jin and **Wentao Zhang**, "Influence of Magnet Layer Numbers on Electromagnetic Performance of Interior Permanent Magnet Machines," in *Proc. Int. Conf. Elect. Mach. Syst.*, Zhuhai, China, 2023, pp. 696–700.

[C14] X. Chen, H. Hua, D. Li, **Wentao Zhang**, Z. Wu and W. Hua, "Modular Stator Surface-Mounted Permanent Magnet Machines with Flux Gaps," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, San Francisco, CA, USA, 2023, pp. 1–7.

[C15] X. Yan, Z. Wu, W. Hua, **Wentao Zhang** and M. Cheng, "A Mathematical Model for Wound Field Switched Flux Machine Considering Inductance Harmonics of Field and Armature Windings," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, San Francisco, CA, USA, 2023, pp. 1–6.

[C16] Z. Wu, L. Jin, **Wentao Zhang**, X. Yan, H. Ge, H. Hua, W. Hua and M. Cheng, "A Lumped Parameter Thermal Model for Variable Flux Reluctance Machine Considering Relative Position of DC and AC Windings," in *Proc. Int. Conf. Elect. Mach.*, Torino, Italy, 2024, pp. 1–7.

[C17] Z. Wu, X. Yan, Z. Wang, **Wentao Zhang** and W. Hua, "Quantitative Analysis of the Torque-Speed Curve for a Flat Wire Permanent Magnet Machine Drive System," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, 2025, pp. 40–44.

[C18] Z. Wu, E. Fu, J. Du, **Wentao Zhang** and W. Hua, "Influence of Stator Core on the Vibration Behavior of Permanent Magnet Machines with Hairpin Winding," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, 2025, pp. 34–39.

[C19] Z. Wu, **Wentao Zhang**, Z. Wang, L. Shao, W. Hua and M. Cheng, "Stator Semi-Skewing Method for High-Speed Permanent Magnet Machine," in *Proc. Int. Conf. Elect. Mach. Syst.*, 2025, pp. 693–697.

Patents (9)

[P1] Z. Wu, **Wentao Zhang**, L. Jin, and W. Hua, "Inductive Ripple Voltage Suppression Method Based on Low-Pass Filter Circuit," CN118508832A, 2024.

[P2] Z. Wu, **Wentao Zhang**, L. Jin, and W. Hua, "Disk Brushless Electric Excitation Magnetic Field Modulation Motor With Double-Row Stator Structure," CN118100572A, 2024.

[P3] Z. Wu, **Wentao Zhang**, H. Huang, and W. Hua, "**Method for Constructing a High-Precision Motor Performance Database Based on a Dual-Model Strategy**," CN117972924A, 2024.

[P4] Z. Wu, **Wentao Zhang**, W. Hua, and Z. Zou, "**Method for Establishing General Lumped Parameter Thermal Network Model of Motor**," CN117094088A, 2023.

[P5] Z. Wu, **Wentao Zhang**, L. Jin, and W. Hua, "Method for Analyzing Harmonic Orders of Induced Ripple Voltage of Excitation Winding," CN115642847A, 2023.

[P6] Z. Wu, **Wentao Zhang**, J. Liao, Y. Fan, W. Hua, and M. Cheng, "Excitation Winding Induced Pulse Voltage Suppression Method Based on Harmonic Current Injection," CN115208245A, 2022.

[P7] Z. Wu, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "High-Precision Field Path Coupling Design Method for Brushless Electric Excitation Magnetic Field Modulation Motor," CN114781227A, 2022.

[P8] Z. Wu, **Wentao Zhang**, W. Hua, and Z. Zhu, "Method for Weakening the Induced Pulsating Voltage of a Brushless Electric Excitation Magnetic Field Modulated Motor," CN114189185A, 2022.

[P9] W. Hua, **Wentao Zhang**, Z. Wu, and S. Ding, "Method for Reducing Torque Ripple and Induced Voltage of Electric Excitation Flux Switching Motor," CN111769697A, 2020.

Referees

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